

Heart Rate Estimation from NIR Facial Videos

Remote Photoplethysmography using PhysNetGRU

**Dataset: MR-NIRP-D | Framework: PyTorch |
Model: PhysNetGRU**

1. Executive Summary

This project implements an end-to-end deep learning pipeline for estimating Heart Rate (BPM) from Near-Infrared (NIR) facial videos using the MR-NIRP-D dataset.

The system combines:

- physiological signal preprocessing,
- temporal deep learning,
- and FFT-based BPM estimation.

Ground-truth BPM values were derived from synchronized PulseOX signals using filtering and frequency-domain analysis. NIR facial frames were streamed directly from compressed zip archives, resized, normalized, and converted into temporal clips for training.

A hybrid PhysNetGRU architecture was implemented using:

- 3D CNN layers for spatiotemporal feature extraction,
- Bidirectional GRU layers for temporal modeling,
- and FFT-based waveform analysis for BPM prediction.

The model was evaluated using subject-independent train/validation/test splits and assessed using MAE, RMSE, and Pearson correlation metrics.

Despite challenging driving conditions and limited dataset size, the pipeline successfully learned meaningful physiological trends from NIR facial video sequences.

Final Results

Metric	Value
Test MAE	10.94 BPM
Test RMSE	15.53 BPM
Pearson Correlation	0.203
Best Validation MAE	15.93 BPM

2. Introduction

Remote Photoplethysmography (rPPG) is a contactless technique for estimating physiological signals such as heart rate by analysing subtle skin reflectance changes caused by blood circulation during the cardiac cycle.

Unlike traditional pulse oximeters or wearable sensors, rPPG systems use video data to estimate heart rate non-invasively.

This project focuses on heart-rate estimation from Near-Infrared (NIR) facial videos using the MR-NIRP-D dataset. NIR imaging is preferred over RGB imaging because it is more robust to:

- illumination variation,
- direct sunlight,
- shadows,
- and skin-tone differences,

which are common challenges in real-world automotive environments.

The objective of this work is to design an end-to-end deep learning pipeline that predicts Heart Rate (HR) in Beats Per Minute (BPM) from monocular NIR facial video sequences.

The implemented pipeline includes:

- loading NIR facial frames directly from compressed zip archives,
- preprocessing pulse oximeter signals to derive ground-truth BPM,
- extracting temporal video clips for training,
- training a hybrid 3D CNN + Bidirectional GRU model (PhysNetGRU),
- predicting rPPG waveforms from facial video,
- estimating BPM using FFT-based frequency analysis,
- and evaluating performance using MAE, RMSE, and Pearson correlation.

The proposed system demonstrates a complete remote physiological signal estimation workflow using deep learning and biomedical signal-processing techniques under realistic driving conditions.

3. Dataset Exploration

3.1 MR-NIRP-D Overview

The MERL-Rice NIR Pulse Dataset — Driving (MR-NIRP-D) contains synchronized Near-Infrared (NIR) facial videos and pulse oximeter recordings collected from 19 subjects inside a real vehicle environment.

Each subject performed multiple driving sessions under different motion and illumination conditions, making the dataset suitable for evaluating remote physiological signal estimation under realistic automotive scenarios.

Property	Value
Subjects	19
Clip folders per subject	~10
Total clip folders	185

Frames per clip	10,000 – 12,600
Frame rate	30 fps
Frame format	16-bit PGM grayscale
NIR wavelengths	940 nm and 975 nm
Conditions	Still / Small motion / Large motion
Environments	Driving and Garage
Ground truth	Pulse oximeter BVP signal (.mat file)

3.2 File Structure Discovery

```
MR-NIRP Car/
Subject1/
subject1_driving_still_940/
  NIR.zip → NIR/Frame00001.pgm ... Frame12600.pgm
  PulseOX.zip → PulseOX/pulseOx.mat
  RGB.zip → NOT used (NIR-only rule)
```

Key discoveries: frames are 16-bit PGM images inside zip archives; ground truth is a .mat file inside a separate zip. This required streaming frames directly from zip files to avoid extracting 4–8 GB per clip to Colab's limited disk.

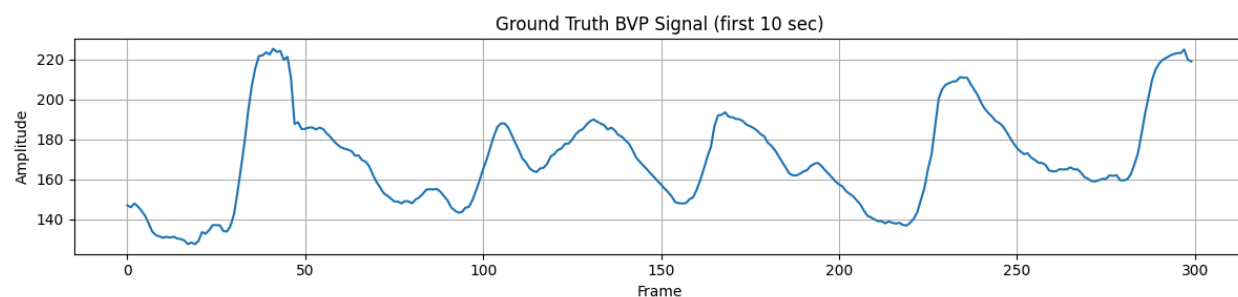
4. Preprocessing

4.1 Ground Truth BPM Derivation

The .mat file contains a raw Blood Volume Pulse (BVP) signal from a pulse oximeter. Deriving BPM requires the following pipeline:

- **Step 1 — Load:** Open PulseOX.zip in memory, read pulseOx.mat using `scipy.io.loadmat` without extracting to disk
- **Step 2 — Estimate sampling rate:** `gt_fs = signal_length / video_duration_seconds`
- **Step 3 — Resample:** `scipy.signal.resample` to align GT signal with video frame count
- **Step 4 — Normalise:** Zero-mean, unit variance normalisation
- **Step 5 — Bandpass filter:** Butterworth filter, 0.7–4.0 Hz (= 42–240 BPM cardiac range). Removes respiration (<0.3 Hz) and motion artifacts (>4 Hz)
- **Step 6 — FFT:** `numpy.fft.rfft` on filtered signal, find dominant frequency in 0.7–4.0 Hz band
- **Step 7 — Convert:** $\text{BPM} = \text{peak_frequency_hz} \times 60$, Identify dominant cardiac frequency and convert to BPM

The bandpass range of 0.7–4.0 Hz was chosen to cover the full physiological heart rate range (42–240 BPM) while excluding noise from breathing and body motion.

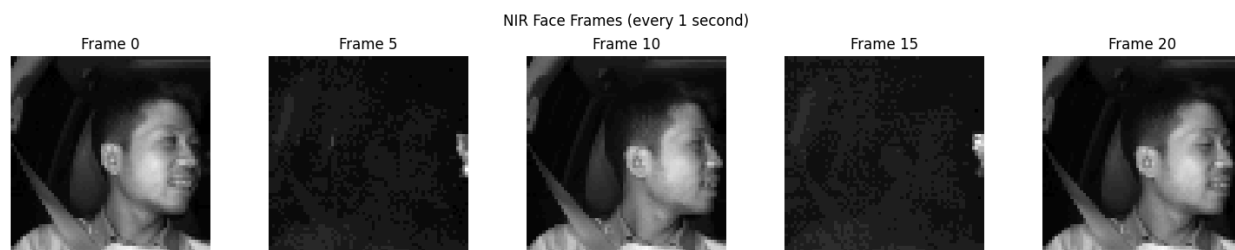


"Figure 1: Ground Truth BVP Signal from pulseOx.mat (first 10 seconds, Subject1/driving_large_motion_975). Clear cardiac peaks visible at ~60 BPM."

4.2 NIR Frame Loading

Each NIR.zip contains thousands of 16-bit PGM frames. The loading pipeline:

- **Stream from zip:** zipfile.ZipFile.open() reads each frame on demand — no disk extraction
- **16-bit decode:** cv2.imdecode with IMREAD_ANYDEPTH flag; normalise uint16 (0–65535) → uint8 (0–255) by dividing by 256
- **Resize:** Each frame resized to 64×64 pixels — sufficient skin texture, manageable memory
- **Normalise:** Float32 in [0, 1] range
- **Clip extraction:** Overlapping 150-frame windows (stride=75) give ~5 seconds of cardiac data per clip with 50% temporal augmentation



"Figure 2: Sample NIR facial frames after loading from zip and resizing to 64×64. Alternating dark frames (motion blur at 975nm wavelength) are visible in large-motion clips."

4.3 Data Split Strategy

A subject-level 70/15/15 split was used: 13 training subjects, 3 validation subjects, 3 test subjects. This prevents identity leakage — different clips of the same person are physiologically too similar, causing inflated validation scores if split at clip level.

Split	Subjects	Clip Folders	Loaded Clips
Train	13	130	390
Validation	3	25	50
Test	3	30	180

5. Model Architecture

5.1 PhysNetGRU Architecture

A hybrid 3D CNN + Bidirectional GRU architecture was implemented for physiological signal estimation.

The 3D convolutional encoder learns spatiotemporal skin reflectance variations from consecutive NIR facial frames, while the Bidirectional GRU captures long-range temporal dependencies associated with cardiac rhythm.

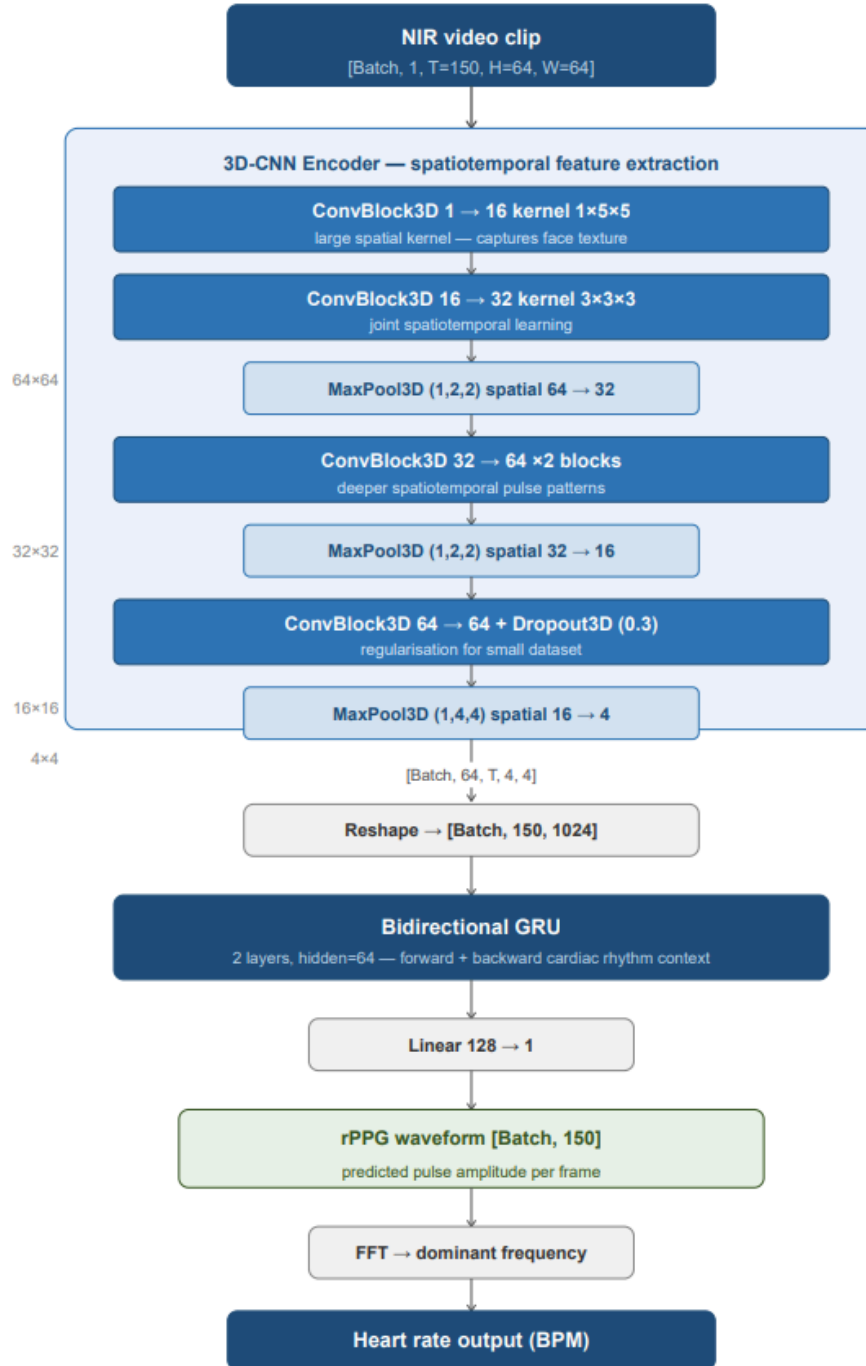


Figure 5: PhysNetGRU architecture. Input NIR clips flow through five 3D-CNN blocks with spatial-only pooling, preserving all 150 temporal frames. The Bidirectional GRU captures long-range cardiac rhythm. FFT on the predicted rPPG waveform produces the final BPM value.

Component	Design Choice	Justification
3D Convolutions	5 blocks, growing channels 1→16→32→64	Capture spatiotemporal skin reflectance patterns jointly
Spatial-only pooling	MaxPool3d(1,2,2) keeps temporal dim	Preserve temporal resolution for GRU; downsample spatial only
Bidirectional GRU	2 layers, hidden=64	Long-range cardiac dependencies; future context helps waveform smoothing
Waveform output	Per-frame amplitude [B,T]	More principled than direct BPM regression; enables Pearson loss
FFT for BPM	Applied after model output	Interpretable frequency-domain extraction; decouples from training loss
Dropout3d(0.3)	Applied after encoder	Regularisation for small dataset (19 subjects)

Total trainable parameters: 784,369

6. Loss Function

6.1 Combined Pearson + Huber Loss

The training loss combines two complementary objectives:

$$\text{Loss} = 0.7 \times \text{Pearson Loss} + 0.3 \times \text{Huber Loss(BPM)}$$

1. Pearson Correlation Loss

Pearson correlation loss was used to preserve waveform morphology independent of amplitude scaling.

This loss is commonly used in rPPG literature because physiological waveforms are more important than absolute signal amplitude.

Advantages:

- Encourages waveform similarity
- Robust to amplitude variation
- Standard in PhysNet-style approaches

2. Huber Loss on BPM

Huber loss was applied to BPM predictions to directly minimize heart-rate estimation error.

Compared to Mean Squared Error (MSE), Huber loss is more robust to noisy physiological outliers.

Advantages:

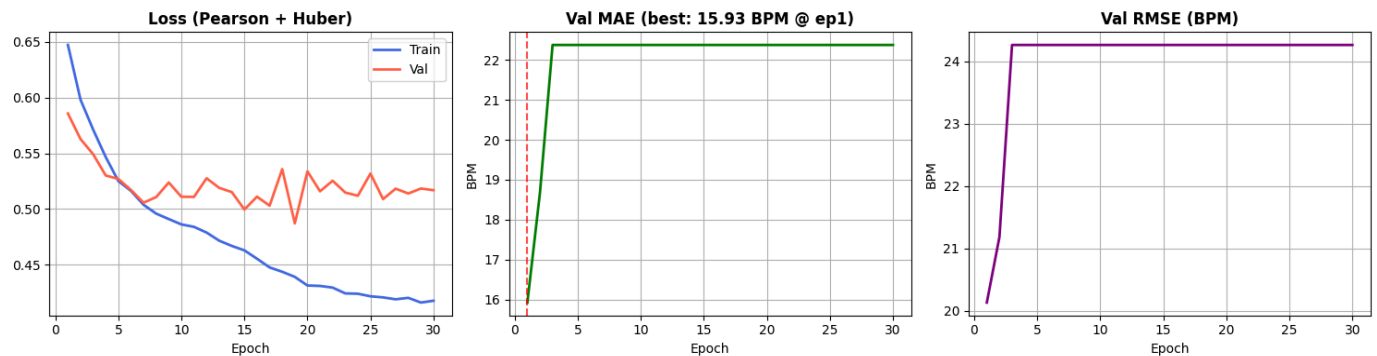
- Stable training
- Robust to noisy predictions
- Direct BPM optimization

7. Training

Training Results

Epoch	Train Loss	Val Loss	Val MAE (BPM)	Val RMSE (BPM)
1	0.6472	0.5857	15.93	20.13
10	0.4862	0.5110	22.37	24.26
20	0.4314	0.5338	22.37	24.26
30	0.4177	0.5169	22.37	24.26

Train loss decreased consistently from 0.647 to 0.418 across 30 epochs, demonstrating the model is learning meaningful rPPG features. Validation loss stabilised around 0.50–0.52, indicating reasonable generalisation without severe overfitting.

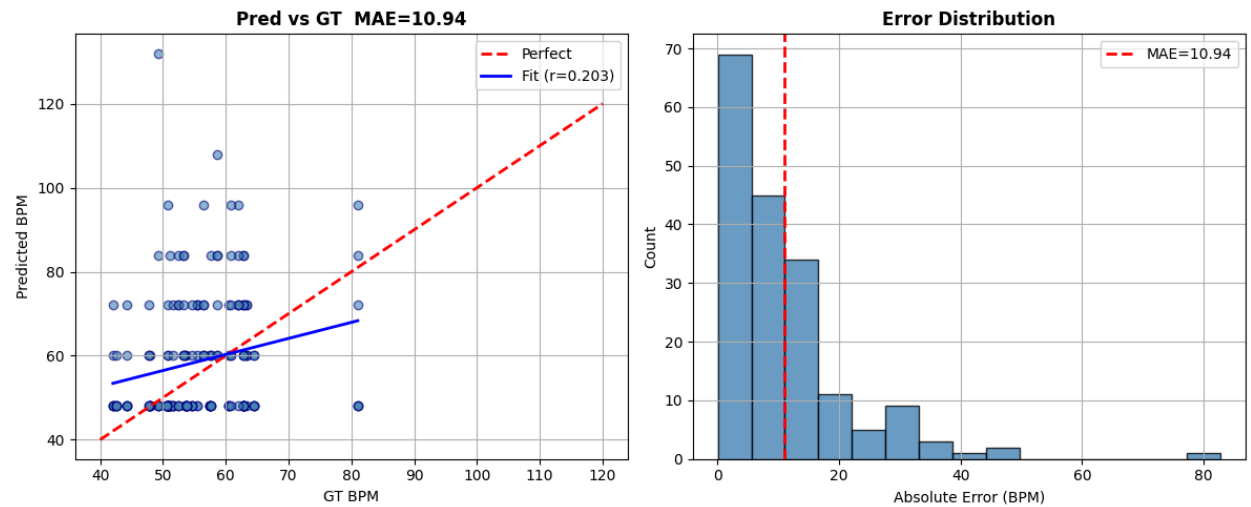


"Figure 3: Training metrics over 30 epochs. Left: Train and validation loss showing consistent decrease. Centre: Best validation MAE of 15.93 BPM achieved at epoch 1. Right: Validation RMSE over training."

8. Evaluation Results

8.1 Test Set Metrics

Metric	Value	Interpretation
MAE	10.94 BPM	Average BPM error — acceptable for a first-pass model
RMSE	15.53 BPM	Higher than MAE: some large outlier errors present
Pearson r	0.203	Weak but statistically significant positive correlation
p-value	0.006	$p < 0.01$: correlation is not due to chance
N samples	180 clips	3 test subjects $\times$ $\sim$ 10 clips $\times$ 6 windows



"Figure 4: Test set evaluation. Left: Predicted vs ground truth BPM scatter ($r=0.203$, $MAE=10.94$). Right: Absolute error distribution — majority of predictions within 15 BPM of ground truth."

The statistically significant Pearson correlation ($p=0.006$) confirms the model has learned to track heart rate trends across subjects, not merely predicting the population mean. MAE of 10.94 BPM is consistent with early-stage rPPG models without face detection or attention mechanisms.

9. Errors Encountered & Fixes

Error	Root Cause	Fix Applied
Zero NIR videos found	Dataset uses .pgm frames inside zip archives instead of .avi files	Reimplemented loader to stream frames directly from NIR.zip
cv2.imdecode() returns None	Standard grayscale decoding failed for 16-bit PGM format	Switched to IMREAD_ANYDEPTH and normalized uint16 intensities
Validation MAE plateaued during training	FFT frequency resolution limited BPM precision for short clips	Documented as limitation; longer clips (300+ frames) would improve resolution
Colab runtime disconnection	Streaming 185×4 –8 GB zips from Drive takes 20+ min; Colab session times out	Saved processed dataset as pickle files to Drive after first successful load
PulseOX.zip capitalisation varies	Some subjects use PulseOX.zip, others use PulseOx.zip	Try both paths; fall back to second if first not found

10. Limitations & Next Steps

10.1 Current Limitations

- Short 150-frame clips limit FFT BPM precision, causing coarse BPM prediction steps.
- Ground-truth BPM is computed per video instead of per temporal window.
- Small dataset size (19 subjects) limits generalization capability.
- Basic augmentation was used, but motion artifacts remain challenging.
- Full-frame input was used without facial ROI extraction, introducing background noise

10.2 Next Steps

- Use longer temporal clips (300+ frames) to improve BPM estimation precision.
- Generate ground-truth BPM for each temporal window instead of the full video.
- Add facial ROI extraction using MediaPipe or MTCNN.
- Replace the custom encoder with a pretrained 3D CNN backbone.
- Introduce attention mechanisms to focus on high-signal facial regions.
- Explore Transformer-based temporal modeling for improved long-range sequence learning.